Network Threats and Physical Security Assignment

1. Common Network Threats

Threat	Example
Phishing	Fake Instagram login page stealing credentials
DDoS	Overloading a food delivery service like Zomato with traffic
Man-in-the-Middle	Intercepting public Wi■Fi traffic used for WhatsApp Web login
Malware	Malicious APK pretending to be Paytm
Password Attack	Brute-force attempts against Instagram accounts

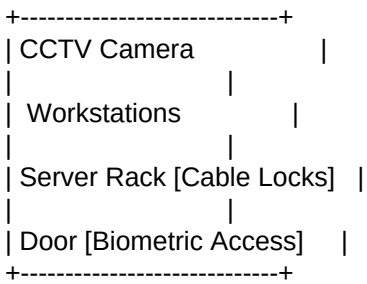
2. Python Brute-Force PIN Simulation

```
preset_pin = "1234"
attempts = 0
for i in range(10000):
    attempts += 1
    guess = f"{i:04d}"
    if guess == preset_pin:
        print("PIN Found:", guess)
        print("Attempts:", attempts)
        break
```

3. Physical Security Risks and Mitigation

Risk	Mitigation
Unauthorized Entry	Biometric access control
Fire	Fire suppression system
Power Failure	UPS and generator backup
Theft of Equipment	Locked racks and CCTV monitoring

4. Small Office Floor Plan



5. College Computer Lab Security Checklist

- Install and regularly update antivirus software on all lab PCs.
- Use strong passwords and multi-factor authentication for admin accounts.
- Restrict student access to network equipment and server rooms.
- Enable firewall protection on all systems.
- Monitor CCTV cameras covering entrances and equipment areas.
- Perform regular data backups and recovery testing.
- Lock network racks and secure cables to prevent tampering.